

AHNAF ZAREEF

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EDUCATION

McMaster University

Bachelor of Computer Engineering

Hamilton, ON

Sep. 2024 – April 2028

Coursework: Digital Systems Design, Electronic Devices and Circuits, Signals & Systems, Microprocessor Systems

EXPERIENCE

Low Voltage Team Member

Sep. 2025 – Present

McMaster Solar Car Project

Hamilton, ON

- Designed IV Curve Tracer PCB in **Altium** using a Darlington pair for high-gain current amplification, **ESP32** and a Hall effect sensor to characterize per-cell efficiency and detect faults.
- Developed protection circuitry: **MOSFET switching** on cell output, and Zener clamping on the ADC for overvoltage.
- Designing the **battery management system (BMS)** emulation schematic in **Altium** for an **LV fault injection** test board, researching and selecting components to emulate BMS behaviour on a low-voltage PCB.
- Mapping potentiometer inputs to simulated pack voltage, current, and temperature over **CAN** to validate **BMS fault thresholds** in connected boards.

Firmware Intern

May 2026 – Sep. 2026

Carobot Learning & Research Organization

Toronto, ON

- Built firmware and host tooling for on-device image classification on an **ESP32-S3**, running inference fully offline.
- Built custom MicroPython firmware on **Linux (WSL)** with ESP-IDF, adding C modules for live camera capture while receiving weights over USB.
- Debugged training stalls during streaming by splitting **MicroPython** and UVC capture into separate **FreeRTOS** tasks.
- Designed schematic and **PCB** in **KiCad** with USB-C and DC/DC power, a motor driver, an **I2S** microphone, and expansion headers for add-on modules.
- Evaluated microcontroller variants, ruling out models for limited **PSRAM** and USB support, and benchmarked MobileNet V1 at **2.1x** available device **RAM**, pivoting to smaller models trained on raw images to fit on-device.

PROJECTS

Neural Network Hardware Accelerator

- Built a binarized neural network accelerator on a **Xilinx Arty S7-25 FPGA**, packaged as **AXI4-Lite IP** in **Vivado** and controlled by a **MicroBlaze** processor, classifying touchscreen input in under **60 μ s**.
- Designed layer datapath and **FSM** in **SystemVerilog** where one neuron is computed sequentially, reusing one layer module for all 3 layers, taking only **7% of FPGA's Logic**.
- Inferred weights as **Block Ram**, delayed inputs one cycle to match BRAM latency and **synchronized** reset button with a 2-flip-flop synchronizer, meeting **100 MHz timing**.

I2C Master Controller

- Designed a **Verilog** I2C master controller to interface a processor with sensors and peripherals over the I2C bus.
- Implemented open-drain signaling and start/stop framing so devices safely share the bus without conflicts.
- Developed a **Verilog testbench** and **slave model** to validate edge cases and open drain bus conflict prior to synthesis on the **Arty S7-25**.

LCD SPI Driver

- Wrote a C driver for an ILI9341 SPI display on a **MicroBlaze** soft-core CPU, communicating over **AXI SPI** and **GPIO** and fitting the entire system in on-chip **BRAM** to run without external memory.
- Built a graphics library for the display, supporting shape fills, text, and images without **framebuffer** to fit on-chip memory.

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, TCL, C, C++, Python, Bash

Protocols: I2C, SPI, UART, AXI, CAN, USB

Tools: Quartus, Vivado, Altium, ESP-IDF, LTSpice, MATLAB, Oscilloscope, Logic Analyzer, Multimeter, Bench Power Supply

Platforms: Xilinx, ARM Cortex, ESP32, STM32, DE10-Lite, FreeRTOS, Linux WSL, Windows CLI